

Roll No.

Total Pages : 2

CMCA/M-22

24529

MOBILE APPLICATION DEVELOPMENT

Paper–MCA-20-43

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. (a) Describe DVM.
- (b) What are the permissions required for setting up the Google maps app in AndroidManifest.xml file?
- (c) Describe the services callbacks and override in an android applications.
- (d) Describe the term near field communication.
- (e) List any three ways to storing data in an android applications. 5×3=15

UNIT-I

2. Describe Android platform architecture. Discuss the various layers and their components and functions. 15
3. (a) Describe the main properties of Linear Layout. 5
- (b) What are the various application components in android? 10

UNIT-II

4. (a) Define the procedure to navigate between Activities and exchange data between them. 8
- (b) Write down the steps for getting API key for Google Maps. 7
5. Differentiate the following with examples :
 - (a) List view and Spinner view.
 - (b) Implicit and Explicit Intent. 15

UNIT-III

6. What are the fragments? How it is different from activities? Write a procedure to add fragment, remove fragment and replace fragment using Fragment transaction. 15
7. (a) Write the use of Media controller class. 5
- (b) Explain the purpose of motion, position and environment sensors with one real-life use case in each. 10

UNIT-IV

8. (a) What is canvas? How to draw a circle using android canvas? 10
- (b) Explain the steps to create a SQLite database in an app. 5
9. (a) How to store data locally/internally in an Android app? Explain. 10
- (b) Write a short note on content provider. 5